

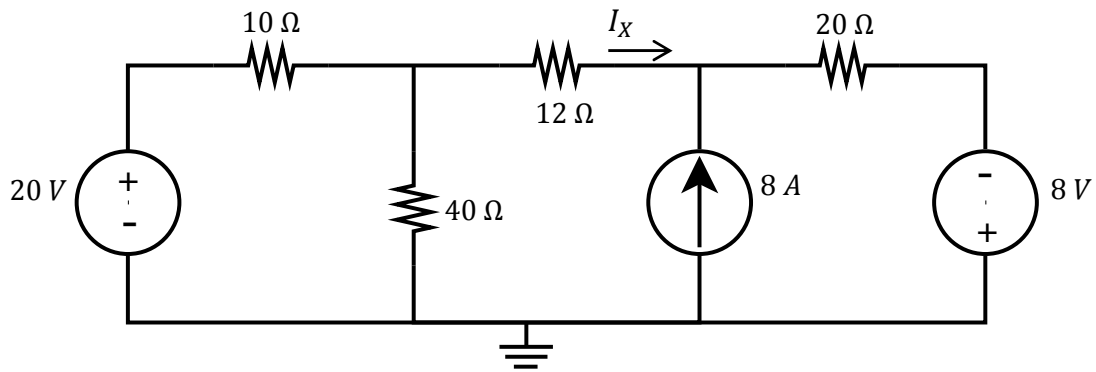
EBN111 KLASTOETS/CLASS TEST 2 (17 APRIL 2014)

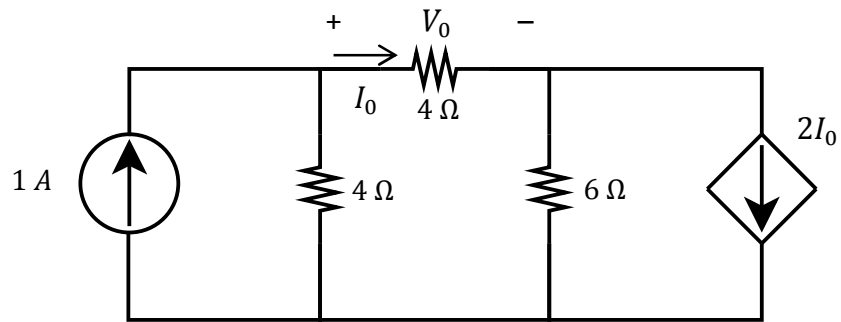
Vraag 1 [2]

Maak gebruik van superposisie en bepaal die bydrae tot I_X slegs as gevolg van die 8 A stroombron.

Question 1 [2]

Use superposition and determine the contribution to I_X by the 8 A current source only.



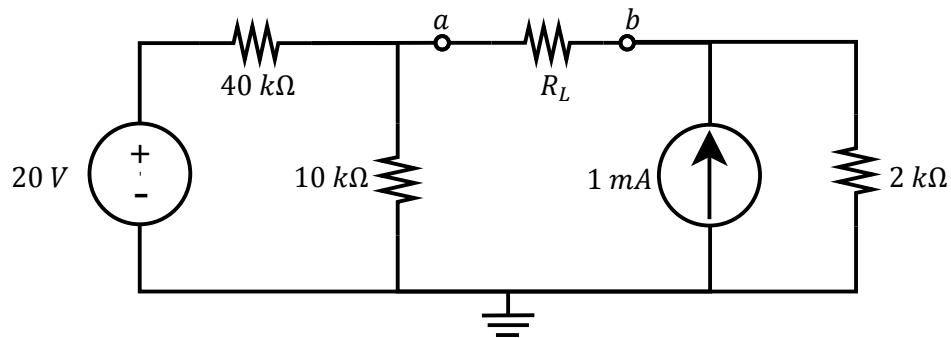
Vraag 2 [2]Maak gebruik van brontransformatie en bepaal V_0 .**Question 2 [2]**Make use of source transformation and determine V_0 .

Vraag 3 [2]

'n Lasweerstand R_L word gekoppel tussen terminale a en b . Die waarde van die lasweerstand moet $10\text{ k}\Omega$ wees vir maksimum drywingsoordrag. Bepaal die Thevenin ekwivalente spanning V_{TH} tov terminale a - b .

Question 3 [2]

A load resistance R_L is coupled between terminals a and b . The value of the load resistance must be $10\text{ k}\Omega$ for maximum power transfer. Determine the Thevenin equivalent voltage V_{TH} wrt terminals a - b .

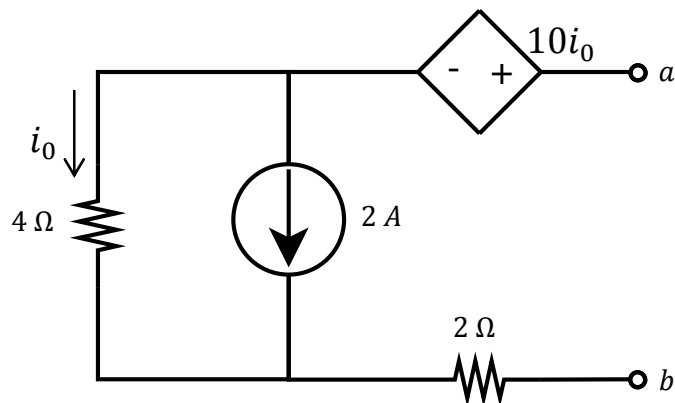


Vraag 4 [2]

Bepaal die Norton ekwivalente weerstand R_N tov terminale a - b .

Question 4 [2]

Determine the Norton equivalent resistance R_N wrt terminals a - b .



Vraag 5 [3]

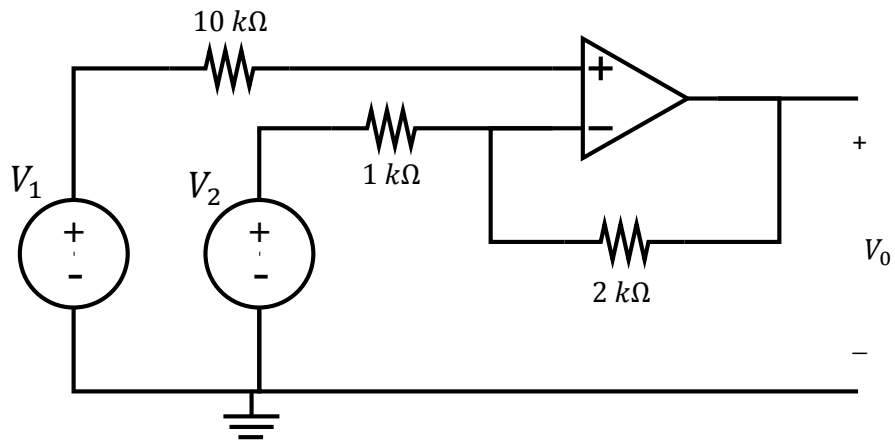
Aanvaar die operasionele versterker is ideaal.

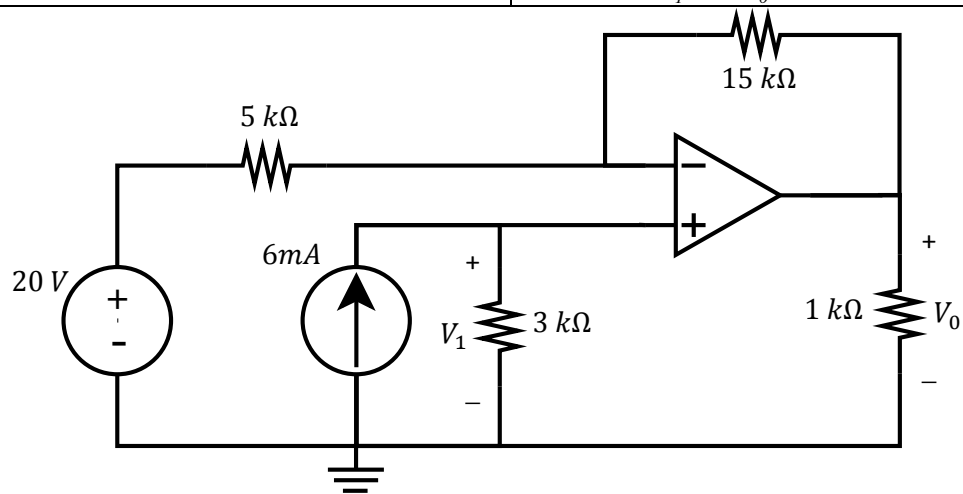
- a) Indien $V_1 = 0$ V en $V_2 = 1$ V, bepaal V_0 .
- b) Indien $V_1 = 1$ V en $V_2 = 0$ V, bepaal V_0 .
- c) Indien $V_1 = 1$ V en $V_2 = 1$ V, bepaal V_0 .

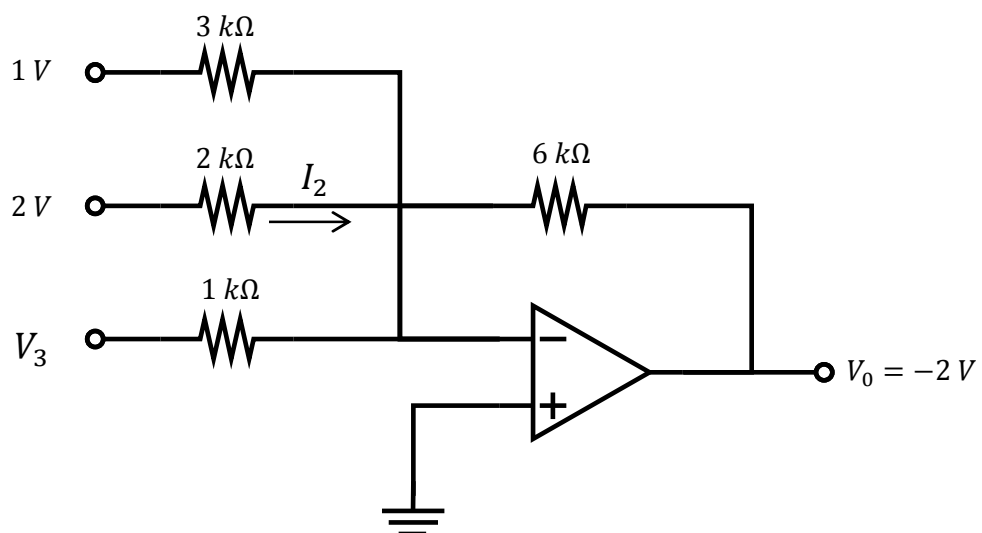
Question 5 [3]

Assume the op-amp is ideal.

- a) If $V_1 = 0$ V and $V_2 = 1$ V, determine V_0 .
- b) If $V_1 = 1$ V and $V_2 = 0$ V, determine V_0 .
- c) If $V_1 = 1$ V and $V_2 = 1$ V, determine V_0 .



Vraag 6 [2]Bepaal V_I en V_O .**Question 6 [2]**Determine V_I and V_O .

Vraag 7 [2]Bepaal I_2 en V_3 .**Question 7 [2]**Determine I_2 and V_3 .

EBN111 KLASTOETS/CLASS TEST 2 (17 APRIL 2014)
ANTWOORDBLAD / ANSWER SHEET

Van / Surname:		Studentenommer: Student number:							
Voorletters / Initials:		Taal / Language:	Afr			Eng			

Tot:	/15
-------------	------------

1	$I_{X(8A)} =$		
2	$V_0 =$		
3	$V_{TH} =$		
4	$R_N =$		
5	$a) V_0 =$	$b) V_0 =$	$c) V_0 =$
6	$V_I =$	$V_0 =$	
7	$I_2 =$	$V_3 =$	